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In a cavern, in a mountain

Canada's only underground ore mill opens

A complete copper crushing and concentrating mill will be officially opened on Vancouver Island today — 8,000 feet inside a mountain — by Mines Minister Kiernan.

The underground mill, the first in Canada, will concentrate the copper ore of the Sunro property of Cowichan Copper Co. It is located at River Jordan on the coast of Vancouver Island, 40 miles west of Victoria.

Cowichan has a four-year sales contract with Japanese interests who will buy all production. Estimated gross value is $16 to $20 million.

The mill is the brainchild of O. G. "Ozzie" MacDonald, president of Cowichan. It has been built at an approximate cost of $1 million, comparable to the usual cost of a mill of its size but only half the cost of building such a mill at the Sunro mine portal.

The two problems facing a copper mill at the Sunro portal are the difficulty of obtaining a firm foundation and the length of mine tunnel — almost a mile and a half — through which the ore would have to be hauled to the mill.

The underground installation consists first of a 72-ton jaw crusher, which takes the broken ore and reduces it to seven-inch thickness, then a gyratory crusher that will reduce to less than three inches and a cone crusher that will produce three-quarter-inch fine ore.

The fine ore will be fed into one of two grinding circuits, each with one rod mill and two ball mills to further grind the ore. The ore then goes to a conditioner, which mixes it with the necessary reagents for flotation.

From the conditioner the ore feed proceeds to the flotation cells, where the concentrate is taken off, dewatered in passage through a thickener and filter, and loaded into five-ton steel containers which are carried on rail cars to the portal, then covered and loaded on trucks for hauling 54 miles to Cowichan's wharf at Hatch Point, on Swartz Bay, for loading onto freighters.

The concentrator waste, or tailings, are pumped through a six-inch plastic pipe to the portal, then a further 5,000 feet to tidewater for dumping.

The underground operation, according to Mr. MacDonald, is an unqualified success. It began production in May, handling 400 tons of ore per day, and has increased steadily up to the present level of 1,000 tons. Its eventual capacity, which will be reached within a few months, will be 1,500 tons per day.

Operating conditions underground are actually superior to those in most outside mills. The temperature is a constant 60 to 65 degrees all year, lighting is provided by two banks of double fluorescent lights for the entire 200-foot mill length, and ventilation and dust removal are provided by pumps from the portal and a ventilation raise.

Noise is less than anticipated, as the rock walls have proved more absorbent than the usual wood or metal. The mill area is completely dry, and rock conditions are good, with the roof bolted every five feet.

There are savings all through the system of ore and concentrate handling. If the ore was trammed out to a mill at the portal, Mr. MacDonald estimates, the cost of ore chutes and proper equipment would be $180,000, and the operational cost 60 cents per ton.

With the mill inside the mine, the tailings pipe cost $60,000, the transportation cost of tailings is 5 cents per ton, and the cost of moving the concentrates to the portal is negligible. The net saving of 50 cents per ton, Mr. MacDonald points out, is equal to the profit on many mine operations.

In addition, the use of the covered concentrate containers for shipping has eliminated a 3 to 5 per cent spill and dust loss, at a saving of $3 to $5 per ton of concentrates. Handling at Hatch Point is made easier, as the containers are merely run up under the roof of a storage shed and tripped, and the concentrate forms its own conical pile.

These and other operating efficiencies have lowered production estimates from $5.78 per ton to $5.05. Preliminary estimates of ore in the River zone, the first to be mined, showed 974,000 tons averaging 1.8 per cent copper. New estimates indicate reserves of 1,947,253 tons of 1.54 per cent ore, and inferred reserves of 772,560 tons of 1.14 per cent.

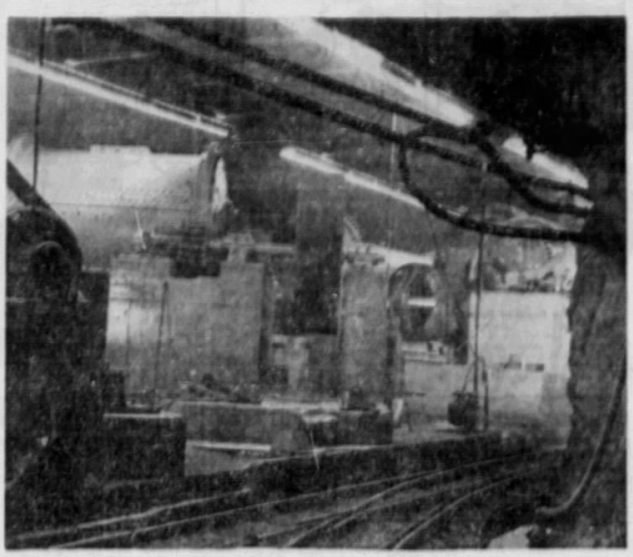
In addition, there are a number of smaller ore bodies that have not yet been explored even as to size, that could prove commercially feasible in the future, says Mr. MacDonald.

Copper was first discovered in the area in 1915, and exploration work carried out intermittently since that time. Consolidated Mining and Smelting did the most work on the property, but found production too costly, since it would have to build a deep-sea wharf and breakwater at Port Renfrew at a cost of $3 to $5 million.

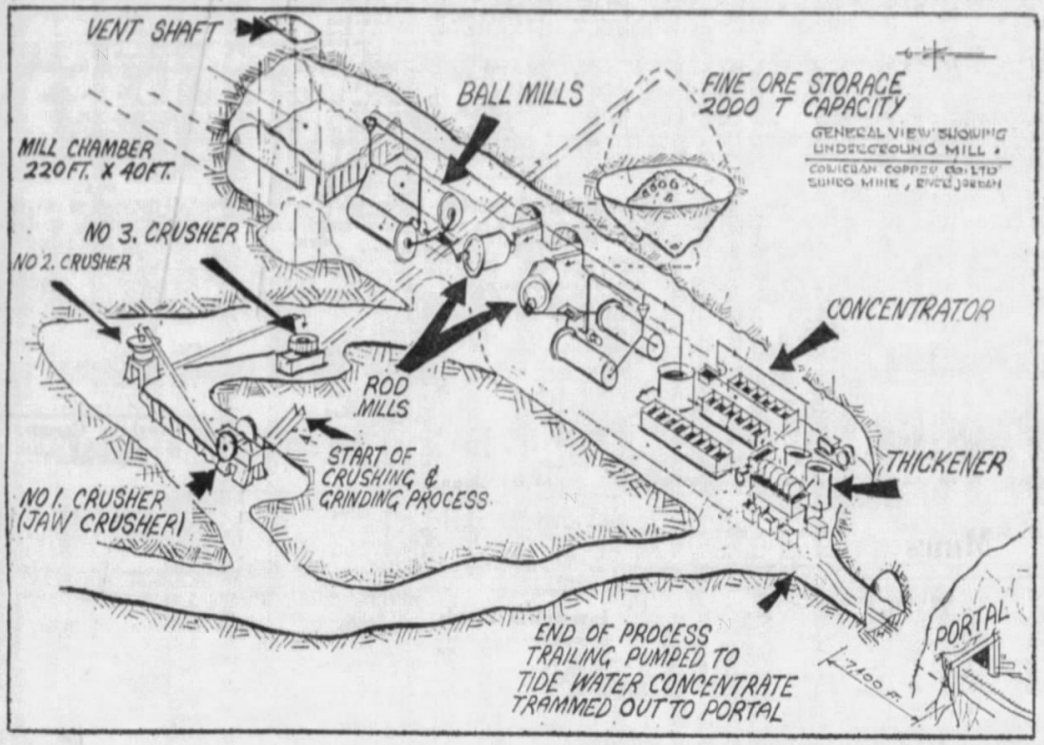
Cowichan has evaded the problem by using the wharf at Hatch Point, built previously to ship concentrates for its Blue Grouse Mine near Cowichan Lake.

The property has been leased for a ten-year period ending in 1970, with two optional five-year extensions, from Sunro Mines Ltd., in which Cominco has a major interest. Payment is in the form of a royalty per ton of ore milled.

Partial financing of the mines was arranged by a loan from the Mitsui Co. of Japan, which will be paid out of production. Mitsui also has an agreement to purchase copper production at the New York price in U.S. funds.



Huge machinery operates, a mile and a half underground at Sunro mine.



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